

Design and Development of an Online Doctor's Appointment System

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Abstract

The increasing adoption of information and communication technologies offers new opportunities to improve healthcare service delivery, yet many healthcare facilities still rely on manual appointment scheduling methods, leading to long waiting times, scheduling conflicts, inefficient record keeping, and limited communication between patients and healthcare providers. To address these challenges, this project proposes the design and future implementation of a web-based online doctor's appointment system that will streamline interactions among patients, doctors, and administrative staff. The system will allow patients to register online, manage their profiles, view available medical services, and schedule appointments at convenient times, while also supporting virtual consultations through Google Meet integration for remote access. It will operate using three main user roles: patient, doctor, and admin enabling doctors to manage their availability and appointments and administrators to oversee appointment approvals and system management. Developed with Next.js and TypeScript for the frontend, Node.js and Express.js for the backend, and PostgreSQL for secure data storage, the system will implement authentication via JSON Web Tokens and secure integration with Google services through OAuth 2.0. With a focus on usability, security, scalability, and performance, the system is expected to reduce administrative workload, enhance patient engagement, and improve communication between patients and healthcare providers, ultimately contributing to more efficient healthcare service delivery in the future.

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Chapter 1

Introduction

Healthcare services in many areas face challenges such as limited infrastructure, high patient demand, and reliance on manual appointment scheduling and record-keeping. These traditional methods often lead to long waiting times, scheduling conflicts, incomplete records, and poor communication between patients and healthcare providers. The lack of digital solutions further limits access and efficiency, particularly in remote or underserved areas.

This project proposes the design and future implementation of a web-based online doctor's appointment system. The system will allow patients to register online, manage profiles, view services, and schedule appointments. It will support virtual consultations through Google Meet and include role-based access for patients, doctors, and administrators. The system aims to reduce administrative workload, improve communication, and provide more efficient and accessible healthcare services.

1.1 Background

Healthcare systems in many developing countries face challenges such as limited infrastructure, high patient demand, and inefficient administrative processes. In Ethiopia, these challenges are particularly evident, as many healthcare facilities still rely on manual methods for managing patient appointments and medical records, including paper-based registration and phone calls. Such traditional approaches often lead to long waiting times, scheduling conflicts, incomplete or lost records, and poor communication between patients and healthcare providers. The limited use of digital technologies further restricts the accessibility and quality of healthcare, especially for patients in remote or underserved areas.

The growing need for efficient, timely, and well-organized healthcare services highlights the potential of digital solutions to address these challenges. Web-based appointment systems can streamline patient registration, scheduling, and record management

while improving communication between patients and healthcare providers. By reducing administrative workload, minimizing errors, and providing timely notifications, such systems can enhance patient satisfaction and overall service delivery, even in resource-limited settings.

This project proposes the design and future implementation of a web-based online doctor's appointment system tailored to the Ethiopian healthcare context. The system will enable patients to register online, manage their personal profiles, view available medical services, and schedule appointments efficiently. It will also support virtual consultations through Google Meet integration, improving access for patients who cannot easily visit healthcare facilities. By implementing role-based access for patients, doctors, and administrators, the system aims to enhance appointment management, streamline communication, and contribute to a more organized, accessible, and efficient healthcare service delivery in Ethiopia.

1.2 Statement of the Problem

Despite the critical role of healthcare services, many facilities continue to rely on manual and paper-based processes for patient registration, appointment scheduling, and record management. These traditional methods are often slow, inefficient, and prone to human error, resulting in long waiting times, missed or overlapping appointments, and incomplete patient records. Patients frequently face difficulties in accessing information about available services or doctors, while healthcare staff spend considerable time on administrative tasks that could otherwise be devoted to patient care.

Additionally, the lack of a centralized digital system makes it difficult to coordinate schedules, track patient history, and provide timely notifications or reminders for upcoming consultations. This gap not only affects the efficiency of healthcare providers but also reduces patient satisfaction and engagement, creating challenges for maintaining quality care. The growing demand for healthcare services further exacerbates these issues, making manual systems increasingly inadequate to meet the needs of both patients and providers.

Therefore, there is a pressing need for a reliable, user-friendly, and accessible digital system that can streamline appointment management, improve communication between patients and healthcare staff, and reduce administrative workload.

1.3 Objectives of the Project

1.3.1 General Objective

The general objective of this project is to design and implement a web-based online doctor's appointment system that improves the efficiency, accessibility, and quality of healthcare services by streamlining patient registration, appointment scheduling, and communication between patients and healthcare providers.

1.3.2 Specific Objectives

The specific objectives of the project include:

- To develop an online platform that allows patients to register, manage their profiles, and schedule appointments conveniently.
- To implement role-based access for patients, doctors, and administrators to ensure secure and organized management of appointments and patient data.
- To integrate virtual consultation support through Google Meet to facilitate remote healthcare services.
- To provide automated notifications and reminders for upcoming appointments and follow-ups.
- To display clear information about medical services and fees, helping patients make informed decisions.
- To reduce administrative workload for healthcare staff by providing an efficient digital system for managing appointments and patient records.

1.4 Purpose of the Project

The purpose of this project is to provide a digital solution that simplifies and improves the management of doctor appointments. By implementing a web-based system, patients will have a convenient way to register, schedule, and track their appointments, while healthcare providers will be able to manage schedules, access patient information, and communicate more effectively. The system also aims to support virtual consultations, reducing the need for physical visits and improving access to healthcare services. Overall, the project seeks to enhance the efficiency, accuracy, and quality of healthcare delivery, benefiting both patients and healthcare staff.

1.5 Scope of the Project

The scope of this project focuses on the development and future implementation of a web-based online doctor's appointment system. The main aspects included in the project are:

Patient Management

- Registration of new patients with personal, contact, and basic medical information.
- Profile management, allowing patients to update their information and view appointment history.
- Online appointment scheduling with the ability to select services, doctors, dates, and times.
- Access to virtual consultation links through Google Meet for remote appointments.
- Receiving notifications and reminders for upcoming appointments and follow-ups.
- Providing feedback on services and consultations.

Doctor Management

- Viewing scheduled appointments and patient information related to assigned consultations.
- Managing availability by setting working hours and available time slots for appointments.
- Accessing virtual consultation links for online sessions.
- Monitoring patient attendance and consultation history for better follow-up.

Administrator Management

- Approving, rejecting, or rescheduling appointment requests.
- Managing patient and doctor records within the system.
- Overseeing system operation, ensuring smooth scheduling and communication.
- Managing notifications and reminders to keep patients and doctors informed.

System Features and Limitations

- Support for role-based access control to ensure secure and organized operations.
- Display of service fees and supported payment methods without direct transaction processing.
- Integration with Google Meet for virtual appointments.
- Focused on appointment management; does not include treatment or direct payment processing modules.
- Designed with usability, security, scalability, and performance in mind to handle multiple concurrent users.

Operational Scope

- The system will be accessible via web browsers on desktops and mobile devices.
- It will provide a centralized platform for managing appointments and patient-doctor interactions.
- The system will improve communication between patients, doctors, and administrative staff while reducing administrative workload.

1.6 Methodology

The methodology outlines the approach and techniques that will be used for the design, development, and testing of the web-based online doctor's appointment system.

1.6.1 Data Gathering Techniques

- **Interviews:** Discussions with healthcare staff and administrators to understand current challenges and requirements.
- **Observation:** Reviewing current manual appointment systems to identify inefficiencies and workflow patterns.
- **Literature Review:** Studying existing online appointment systems and related research to incorporate best practices and modern technologies.

1.6.2 Design Methodology

- **Requirement Analysis:** Identifying functional and non-functional requirements for patients, doctors, and administrators.
- **System Modeling:** Developing use case diagrams, class diagrams, sequence diagrams, and activity diagrams.
- **Interface Design:** Creating user-friendly interfaces to ensure accessibility for patients.
- **Role-Based Access Design:** Defining access control for patients, doctors, and administrators.

1.6.3 Implementation Methodology

- **Frontend:** Next.js with TypeScript for a responsive, fast, and secure user interface.
- **Backend:** Node.js with Express.js for handling business logic, APIs, and Google Meet integration.
- **Database:** PostgreSQL for reliable and consistent storage of patient records.
- **Authentication & Security:** JSON Web Tokens (JWT) for secure login sessions, and OAuth 2.0 for integration with Google services.

1.6.4 Testing Methodology

- **Unit Testing:** Testing individual components for correct functionality.
- **Integration Testing:** Ensuring that different modules work together as expected.
- **System Testing:** Conducted in Alpha and Beta phases.

Chapter 2

Requirement Analysis Description

2.1 Overview of the Existing System

Currently, most healthcare facilities rely on manual processes to manage patient appointments and records. Patients typically register in person at the reception desk, where their personal and medical information is recorded in physical logbooks or spreadsheets. Appointment schedules are often maintained manually, creating challenges in coordinating patient visits, avoiding conflicts, and tracking history.

2.1.1 Activities of the System

- **Patient Registration:** Patients provide personal and medical information at the reception.
- **Appointment Scheduling:** Staff manually check doctor's availability and assign time slots.
- **Consultation Management:** Doctors review schedules; notes are on paper.
- **Communication and Notifications:** Confirmations are verbal or via phone.
- **Record Maintenance and Reporting:** Manual maintenance is time-consuming and prone to errors.

2.1.2 Problems of the Existing System

- Long waiting times due to manual registration.
- Scheduling conflicts caused by lack of centralized system.
- Incomplete, lost, or damaged patient records.
- Limited patient access to appointment information.

- Communication gaps; reminders handled verbally.
- High administrative workload for staff.
- Inefficient reporting and monitoring.

2.1.3 Weaknesses and Strengths

Weaknesses

- Time-consuming manual processes.
- Frequent scheduling conflicts without centralized coordination.
- Paper-based records prone to loss or damage.
- Burdened administrative staff and risk of human error.
- Lack of scalability for growing patient volumes.

Strengths

- Simple and familiar to staff; minimal training needed.
- Low initial cost; no digital infrastructure required.
- Operates independently of internet access or electronic devices.
- Direct control over records for small facilities.

2.2 Business Rules

1. Each patient must register with a unique identifier to ensure proper tracking.
2. Appointments requested by patients must be approved or confirmed by the administrator.
3. Administrators have full access to manage patient and doctor records.
4. Virtual consultations are only accessible through automatically generated Google Meet links.
5. Patients are responsible for providing accurate personal and contact information.
6. All appointments, records, and notifications must be stored securely.
7. System notifications and reminders must be sent automatically to patients and doctors.

2.3 Overview of the Proposed System

The proposed system is a web-based online doctor's appointment platform designed to overcome the limitations of the existing manual system.

2.3.1 Key Features

- **Patient Registration and Profile Management:** Accounts with medical history tracking.
- **Role-Based Access Control:** Secure permissions for patients, doctors, and admins.
- **Appointment Scheduling:** Online booking with conflict checks.
- **Virtual Consultation Support:** Google Meet integration for remote access.
- **Automated Notifications:** Reminders for upcoming appointments.
- **Display of Fees:** Information on service costs and payment options.
- **Admin Dashboard:** Centralized control for managing accounts and activity.
- **Patient Portal and Feedback:** Access to history and engagement tools.

2.3.2 Non-Functional Requirements

Usability: Intuitive and user-friendly interface with responsive design.

Security: Protected data with encryption and role-based access.

Reliability and Availability: 24/7 availability with backup mechanisms.

Scalability: Capable of handling multiple concurrent users and growing data.

Performance: Fast response times for maintaining user satisfaction.

Maintainability: Modular and organized design with proper documentation.

Portability: Accessible via different devices and modern browsers.

2.4 System Requirements (Hardware and Software)

Hardware

- Standard computers or laptops with internet access.
- Mobile devices with modern web browsers for patients.
- Server capable of handling simultaneous users.

Software

- **Frontend:** Next.js with TypeScript.
- **Backend:** Node.js with Express.js.
- **Database:** PostgreSQL.
- **Security:** JWT and OAuth 2.0.
- **API:** RESTful services.

2.5 Constraints and Assumptions

Constraints

- No direct online payment processing handling.
- Reliance on stable internet connectivity for virtual consultations.
- Scope limited to appointment management; no diagnosis or prescription modules.

Assumptions

- Users have basic familiarity with web browsers.
- Staff will maintain accurate schedules and information.
- Healthcare facilities provide adequate internet infrastructure.

Chapter 3

System Modeling

System modeling provides a visual representation of how the online doctor's appointment system operates.

3.1 Use Case Model

The use case model represents the interactions between the system and its actors.

3.1.1 Actor Specification

- **Patient:** Registers, schedules appointments, and provides feedback.
- **Doctor:** Manages availability and conducts consultations.
- **Administrator:** Approves appointments and monitors system activity.

3.1.2 Use Case Diagram

- **Patient:** Register, schedule appointment, access virtual consultation, receive notifications, provide feedback.
- **Doctor:** Manage availability, view appointments, conduct consultations.
- **Administrator:** Approve/reject appointments, manage accounts, monitor system operations.

Figure 1 placeholder: Overall Use-Case Diagram

Figure 3.1: Overall Use-Case Diagram

Figure 2 placeholder: Patient Use Case Diagram

Figure 3.2: Patient Use Case Diagram

Figure 3 placeholder: Doctor Use Case Diagram

Figure 3.3: Doctor Use Case Diagram

Figure 4 placeholder: Administrator Use Case Diagram

Figure 3.4: Administrator Use Case Diagram